

Reviewer Pack — Minimal Local Induction Dimension over the Boolean Ring and ...

Entry 39442532134e · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Local Induction Dimension over the Boolean Ring and Communication Complexity Rank

Entry ID: 39442532134e **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-05 01:32:22 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Topology (Local Induction Dimension) **Field B** (complexity object): Communication Complexity (Matrix Rank)

Statement:

For every k -CNF formula φ with n variables, the minimal local induction dimension (Ind) of the associated matroid derived from φ over the Boolean ring is linearly correlated with its communication complexity rank, such that $\text{Ind}(\varphi) = \Theta(\text{rank}(\varphi))$.

Rationale (proposer's reasoning):

Local induction dimension captures the combinatorial complexity of a matroid, and by considering its value over the Boolean ring, we may expose hidden structure in communication complexity. This bridge might reveal inherent relationships between linear algebraic properties of CNFs and their corresponding communication complexity.

Taxonomy category: c003b_cumulative_entropy/SC2#c49 (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 3253b0fa6dc0e952

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, across all k -CNF formulas φ with $n \leq 40$ variables, the ratio of the minimal local induction dimension ($\text{Ind}(\varphi)$) to the communication complexity rank ($\text{rank}(\varphi)$) for each matroid M_φ is bounded by a constant factor of C , where $C \geq 1$. The conjecture is falsified if any instance violates this bound.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.90	SAFE	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "local induction dimension" AND "communication complexity rank" AND boolean ring" - "minimal local induction dimension" site:arxiv.org AND CNF formula" - "matroid derived from CNF" AND boolean ring AND communication complexity

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=5.5s

5.1 Generated Python source

```
import random
from fractions import Fraction

def generate_kcnf(n, k):
    clauses = []
    for _ in range(k):
        clause = set()
        while len(clause) < 2:
            var = random.randint(1, n)
            if var not in clause:
                clause.add(var)
        clauses.append(list(clause))
    return clauses

def gaussian_elimination(A):
    m, n = len(A), len(A[0])
    for pivot_row in range(m):
        pivot_col = next((col for col in range(n) if A[pivot_row][col] != 0), None)
        if pivot_col is None:
            continue
        # Make the pivot element 1
        factor = Fraction(1, A[pivot_row][pivot_col])
        for j in range(n):
            A[pivot_row][j] *= factor
        # Eliminate the pivot column in other rows
        for i in range(m):
            if i != pivot_row and A[i][pivot_col] != 0:
                factor = Fraction(A[i][pivot_col], A[pivot_row][pivot_col])
                for j in range(n):
                    A[i][j] -= factor * A[pivot_row][j]
    return A

def matroid_rank(clauses):
```

```

n = len(clauses)
A = [[0] * (n + 1) for _ in range(n)]
for i, clause in enumerate(clauses):
    for var in clause:
        A[i][var - 1] = 1
A = gaussian_elimination(A)
rank = sum(1 for row in A if any(row[j] != 0 for j in range(n)))
return rank

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = random.randint(5, 40)
    k = random.randint(1, n * (n - 1) // 2)
    clauses = generate_kcnf(n, k)

    try:
        rank = matroid_rank(clauses)
        lnd = len(clauses) # Simplified for this test
        ratio = Fraction(lnd, rank) if rank != 0 else None

        return {
            "metric_name": "lnd_to_rank_ratio",
            "metric_value": float(ratio) if ratio is not None else 0,
            "instances_tested": 1,
            "n_max": n,
            "conjecture_holds": True if ratio is not None and 0.5 <= ratio <= 2 else False,
            "counterexample": "" if ratio is not None and 0.5 <= ratio <= 2 else "lnd_to_rank_ratio_out_of_bounds"
        }
    except Exception as e:
        return {
            "metric_name": "lnd_to_rank_ratio",
            "metric_value": 0,
            "instances_tested": 1,
            "n_max": n,
            "conjecture_holds": False,
            "counterexample": str(e)
        }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if len(sys.argv) > 1 else [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97]
    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    if all(result["conjecture_holds"] for result in results):
        mean_value = sum(result["metric_value"] for result in results) / len(results)
        std_value = (sum((result["metric_value"] - mean_value) ** 2 for result in results) / len(results)) ** 0.5
        support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(not result["conjecture_holds"] for result in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"lnd_to_rank_ratio_out_of_bounds\" first_failing_seed={first_failing_seed}")

```

```

else:
    print("RESULT: INCONCLUSIVE all_trials_used_n_1")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

False, 'counterexample': 'lnd_to_rank_ratio_out_of_bounds'}
TRIAL: {'metric_name': 'lnd_to_rank_ratio', 'metric_value': 1.3333333333333333, 'instances_tested': 1, 'n_max': 7, 'co': 7}
TRIAL: {'metric_name': 'lnd_to_rank_ratio', 'metric_value': 0, 'instances_tested': 1, 'n_max': 7, 'co': 7}
TRIAL: {'metric_name': 'lnd_to_rank_ratio', 'metric_value': 7.52, 'instances_tested': 1, 'n_max': 25, 'co': 25}
TRIAL: {'metric_name': 'lnd_to_rank_ratio', 'metric_value': 3.782608695652174, 'instances_tested': 1, 'n_max': 9, 'co': 9}
TRIAL: {'metric_name': 'lnd_to_rank_ratio', 'metric_value': 0, 'instances_tested': 1, 'n_max': 9, 'co': 9}
TRIAL: {'metric_name': 'lnd_to_rank_ratio', 'metric_value': 4.605263157894737, 'instances_tested': 1, 'n_max': 16, 'co': 16}
TRIAL: {'metric_name': 'lnd_to_rank_ratio', 'metric_value': 4.0, 'instances_tested': 1, 'n_max': 16, 'co': 16}
TRIAL: {'metric_name': 'lnd_to_rank_ratio', 'metric_value': 8.666666666666666, 'instances_tested': 1, 'n_max': 17, 'co': 17}
TRIAL: {'metric_name': 'lnd_to_rank_ratio', 'metric_value': 5.842105263157895, 'instances_tested': 1, 'n_max': 17, 'co': 17}
TRIAL: {'metric_name': 'lnd_to_rank_ratio', 'metric_value': 8.0, 'instances_tested': 1, 'n_max': 17, 'co': 17}
RESULT: FALSIFIED counterexample="lnd_to_rank_ratio_out_of_bounds" first_failing_seed=11

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test code simplifies the minimal local induction dimension (lnd) to the number of clauses, which is not a correct measure for lnd over the Boolean ring. The actual computation of lnd is complex and requires more than just counting clauses.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test results indicate that there exists an instance with $n \leq 40$ variables where the ratio of the minimal local induction dimension (lnd) to the co | next: Investigate the computation of lnd over the Boolean ring and verify the correctness of the test code. If the computation is correct, further analyze the instances where the conjecture fails to identify patterns or conditions that lead to violations of the bound.

11. Audit log (LLM calls)

Total LLM calls: 12

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13734
2	propose	ollama_remote	glm4:latest	0	0	12500
3	propose	ollama_remote	glm4:latest	0	0	12819
4	preregistration	ollama_remote	glm4:latest	0	0	9922
5	novelty	ollama_remote	glm4:latest	0	0	8334
6	novelty	ollama_remote	glm4:latest	0	0	8929
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	19078

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11431
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12850
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11692
11	critic	ollama_remote	glm4:latest	0	0	14287
12	judge	ollama_remote	glm4:latest	0	0	9696

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 145275 ms total latency. Provider mix: {'ollama_remote': 12}

(full prompt+response transcripts available in *research/audit/39442532134e.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/39442532134e.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/39442532134e.tar.gz* (if generated)